

2. [Maximum mark: 5]

(a) State the value of the first term,  $u_1$ . [1]

(b) Given that the  $n^{\text{th}}$  term of this sequence is  $-33$ , find the value of  $n$ . [2]

(c) Find the common difference,  $d$ . [2]

A series of horizontal dotted lines for writing.



Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

## Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 5]

The following table shows the Mathematics test scores ( $x$ ) and the Science test scores ( $y$ ) for a group of eight students.

|                            |    |    |    |    |    |    |    |    |
|----------------------------|----|----|----|----|----|----|----|----|
| Mathematics scores ( $x$ ) | 64 | 68 | 72 | 75 | 80 | 82 | 85 | 86 |
| Science scores ( $y$ )     | 67 | 72 | 77 | 76 | 84 | 83 | 89 | 91 |

The regression line of  $y$  on  $x$  for this data can be written in the form  $y = ax + b$ .

- (a) Find the value of  $a$  and the value of  $b$ . [2]
- (b) Write down the value of the Pearson's product-moment correlation coefficient,  $r$ . [1]
- (c) Use the equation of your regression line to predict the Science test score for a student who has a score of 78 on the Mathematics test. Express your answer to the nearest integer. [2]

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3. [Maximum mark: 5]

Consider an arithmetic sequence where  $u_8 = S_8 = 8$ . Find the value of the first term,  $u_1$ , and the value of the common difference,  $d$ .

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4. [Maximum mark: 6]

Events  $A$  and  $B$  are independent and  $P(A) = 3P(B)$ .

Given that  $P(A \cup B) = 0.68$ , find  $P(B)$ .

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6. [Maximum mark: 6]

Let  $A$  and  $B$  be two independent events such that  $P(A \cap B') = 0.16$  and  $P(A' \cap B) = 0.36$ .

(a) Given that  $P(A \cap B) = x$ , find the value of  $x$ . [4]

(b) Find  $P(A' | B')$ . [2]

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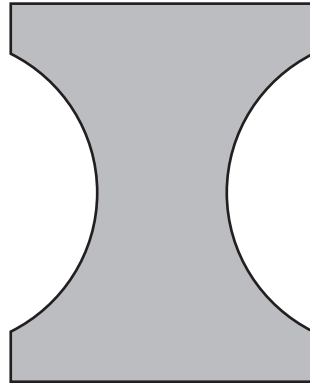
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3. [Maximum mark: 6]

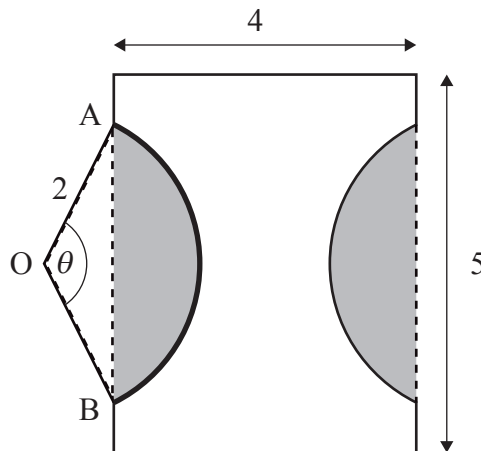
A company is designing a new logo. The logo is created by removing two equal segments from a rectangle, as shown in the following diagram.

diagram not to scale



The rectangle measures 5 cm by 4 cm. The points A and B lie on a circle, with centre O and radius 2 cm, such that  $\angle AOB = \theta$ , where  $0 < \theta < \pi$ . This information is shown in the following diagram.

diagram not to scale



(a) Find the area of one of the shaded segments in terms of  $\theta$ . [3]

(b) Given that the area of the logo is  $13.4 \text{ cm}^2$ , find the value of  $\theta$ . [3]

(This question continues on the following page)



(Question 3 continued)

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12EP05

Turn over

6. [Maximum mark: 7]

Consider the binomial expansion  $(x + 1)^7 = x^7 + ax^6 + bx^5 + 35x^4 + \dots + 1$  where  $x \neq 0$  and  $a, b \in \mathbb{Z}^+$ .

- (a) Show that  $b = 21$ . [2]

The third term in the expansion is the mean of the second term and the fourth term in the expansion.

- (b) Find the possible values of  $x$ . [5]

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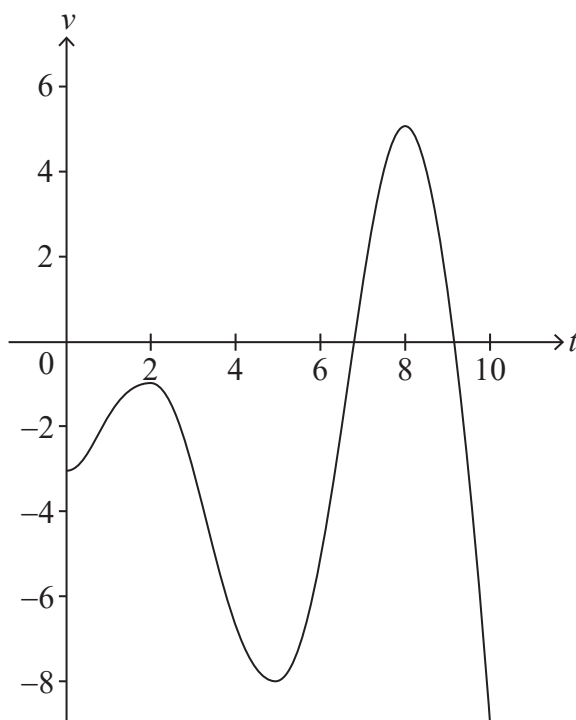




5. [Maximum mark: 6]

A particle moves in a straight line. The velocity,  $v \text{ ms}^{-1}$ , of the particle at time  $t$  seconds is given by  $v(t) = t \sin t - 3$ , for  $0 \leq t \leq 10$ .

The following diagram shows the graph of  $v$ .



- (a) Find the smallest value of  $t$  for which the particle is at rest. [2]
- (b) Find the total distance travelled by the particle. [2]
- (c) Find the acceleration of the particle when  $t = 7$ . [2]

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6. [Maximum mark: 7]

Consider  $f(x) = 4 \sin x + 2.5$  and  $g(x) = 4 \sin\left(x - \frac{3\pi}{2}\right) + 2.5 + q$ , where  $x \in \mathbb{R}$  and  $q > 0$ .

The graph of  $g$  is obtained by two transformations of the graph of  $f$ .

(a) Describe these two transformations.

[2]

The  $y$ -intercept of the graph of  $g$  is at  $(0, r)$ .

(b) Given that  $g(x) \geq 7$ , find the smallest value of  $r$ .

[5]

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